Rule-Based Monetary Policy in China

A Counterfactual Analysis Using the Caravello et al. (2025) Framework

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Motivation

The puzzle. Since 2023, Chinese CPI inflation has hovered near zero, persistently undershooting the official targets set in the annual Government Work Report (GWR).

- GWR CPI targets 2023–2025: **3%, 3%, 2%**
- Realized CPI YoY 2023–2025: roughly **0%**
- Xi's speech at the 2024 Central Economic Work Conference first introduced the phrase "a reasonable rebound in the price level" (物价合理回升); the 2025 CEWC explicitly tied this objective to the conduct of monetary policy.

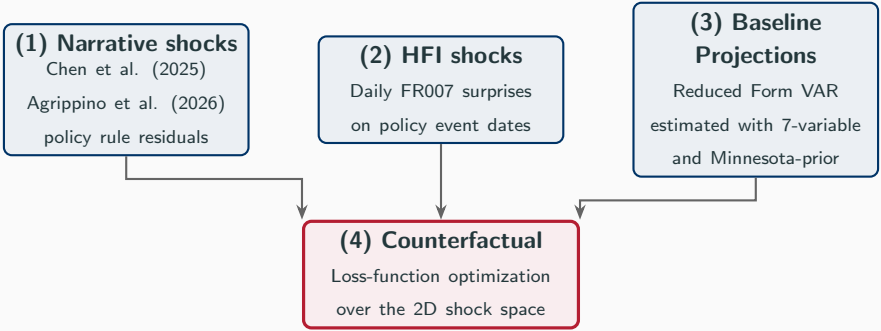
The question. What would have happened to the Chinese economy if the PBoC had strictly followed some form of rule-based policy during this period?

Why it matters.

- Quantifies the policy gap: how far was actual policy from target-consistent policy?
- Tests whether conventional interest rate policy is even capable of closing the inflation gap, given China's transmission mechanism.

Methodology Roadmap

Following Caravello, McKay & Wolf (2025), the empirics-only counterfactual requires **four building blocks**:



Methodology Roadmap Explanation

- Blocks (1) and (2) provide **two distinct empirical IRFs** of the Chinese economy to monetary policy shocks — the policy "treatments" whose linear combinations span the feasible policy space.
- Block (3) provides the **baseline projection** of where the economy goes absent any policy intervention.
- Block (4) solves for the shock sequence that moves the baseline toward alternative paths, subject to a multi-objective loss function.

Narrative Shocks: Estimated Policy Rule

Specification. A monthly policy rule for the 7-day interbank repo rate (FR007), combining elements from Chen, Xiao & Zha (2025) and Miranda-Agrippino, Nenova & Rey (2026):

$$\begin{aligned}\text{FR007}_t \;=\; \alpha \;+\; \rho \cdot \text{FR007}_{t-1} \;+\; \beta_\pi \cdot \text{cpi_gap}_{t-1} \\ +\; \beta_y^+ \cdot \text{gap_pos}_{t-1} \;+\; \beta_y^- \cdot \text{gap_neg}_{t-1} \;+\; \beta_e \cdot \text{fx_gap}_{t-1} \;+\; \varepsilon_t^{\text{MP}}\end{aligned}$$

Variable definitions.

- $\text{cpi_gap}_t = \pi_t - \pi_t^*$, where π_t^* is the annual CPI target from the Government Work Report
- $\text{gdp_gap}_t = g_t - g_t^*$, where g_t is a proxy to monthly real GDP YoY (see appendix: ▸ Stock–Watson construction) and g_t^* is the GWR growth target
- gap_pos_t, gap_neg_t: asymmetric split of the GDP gap around a 1pp threshold (see appendix: ▸ regime-switch motivation)
- fx_gap_t: post-2006 indicator (when fixed exchange rate regime ended) interaction between the change in the central parity rate and the spot–parity deviation

Narrative Shocks: Identification and Rationale

Identification. The residual $\widehat{\varepsilon}_t^{\mathrm{MP}}$ isolates the unsystematic component of FR007 movements — the narrative monetary policy shock series. ▸ actual vs. predicted FR007

Why this hybrid specification?

- CXZ2025 kept FR007 as the dependent variable but excluded the exchange rate; MANR2026 added the foreign exchange term but used the a Composite Monetary Policy Indicator on the left-hand side. ▸ CMPI vs. FR007
- We retain FR007 (the actual market rate that anchors HFI surprises downstream) while adopting MANR2026's FX specification and the lag terms of those gap variables, which mechanically make the policy rates forward-looking.

▸ narrative IRFs

High-Frequency Shocks: FR007 Surprises on Policy Event Dates

Identification strategy. Following Das & Song (2022), we identify monetary policy surprises as the daily close-to-close change in the FR007 rate around dates on which the PBoC announces a change to one of its core policy instruments:

$$\text{shock}_t = \left(\text{FR007}_t - \text{FR007}_{t-1}\right) \cdot \mathbb{1}\{t \in \mathcal{E}\}$$

where $\mathcal{E}$ is the set of policy event dates.

Core policy instruments tracked. Event dates are hand-collected from PBoC:

- 7-day reverse repo rate (OMO) — the primary short-term policy rate post-2016
- 1-year Medium-term Lending Facility (MLF) — introduced 2014, principal medium-term policy tool
- 1-year Loan Prime Rate (LPR) — replaced benchmark lending rate as the loan pricing anchor in August 2019
- Reserve Requirement Ratio for large financial institutions (RRR)
- Benchmark deposit and lending rates (pre-2015 era, before the rate liberalization)

Reduced-Form Long-Term Projection: Bayesian VAR

Specification. A 7-variable monthly BVAR with $p = 6$ lags, estimated with a Minnesota-type prior following Giannone, Lenza & Primiceri (2015):

$$y_t = c + \sum_{\ell=1}^{6} A_\ell \, y_{t-\ell} + \sum_{k=1}^{4} d_k \, D_t^{\text{covid},k} + u_t, \qquad u_t \sim \mathcal{N}(0, \Sigma_u)$$

where $D_t^{\text{covid},k}$ are monthly dummies for Jan–Apr 2020.

Variables in y_t (BVAR ordering, monthly).

$$y_t = \left(g_t,\ \text{IP}_t,\ \pi_t,\ i_t,\ m_t,\ e_t,\ y_t^{US}\right)'$$

g_t: real GDP YoY; IP_t: Industrial Value Added YoY; π_t: CPI YoY; i_t: FR007; m_t: M2 YoY growth; e_t: RMB Real Effective Exchange Rate YoY; y_t^{US}: US output gap YoY; variables reflecting the labor market condition were dropped due to quality and availability issues.

Reduced-Form Long-Term Projection

Estimation samples.

- Monthly estimation with COVID dummies included in all runs.
- A_ℓ estimated for sample period from 2002 to 2022, and 2025.

Baseline projection.

$$\hat{y}_{t^*+h} = \hat{c} + \sum_{\ell=1}^{6} \hat{A}_\ell \, \hat{y}_{t^*+h-\ell}, \qquad h = 1, 2, \ldots, H$$

▸ forecast vs. actual (proxy) monthly GDP

Proposition 1 from Caravello et al.2025 establishes the counterfactual's validity using the Wold representation of the data. Operationally, we follow their approach to compute the baseline path as the BVAR's iterated projection from the pre-counterfactual state, which is equivalent to the Wold representation under invertibility.

Solving the Counterfactual

Inputs. We take three objects from earlier blocks: baseline paths $\pi^{\text{base}}, y^{\text{base}}, i^{\text{base}}$ over a T-month window; transmission maps M_π, M_y, M_i linking narrative and HFI shocks to each variable; and policy targets π^* (GWR CPI target) and y^* (GWR growth target).

Unknown. We solve for $\tilde{\nu} \in \mathbb{R}^{2T}$, containing one narrative shock and one HFI shock each month. Counterfactual paths are baseline plus transmission:

$$\tilde{\pi} = \pi^{\text{base}} + M_\pi \tilde{\nu}, \quad \tilde{y} = y^{\text{base}} + M_y \tilde{\nu}, \quad \tilde{i} = i^{\text{base}} + M_i \tilde{\nu}.$$

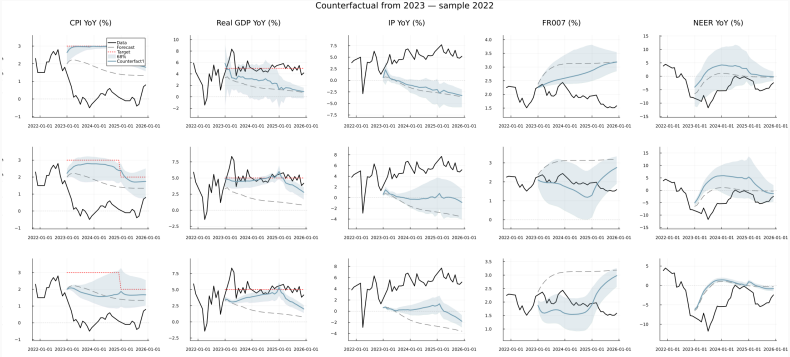
Loss. The shock sequence is chosen to minimize:

$$\mathcal{L}(\tilde{\nu}) = \lambda_\pi \|\tilde{\pi} - \pi^*\|^2 + \lambda_y \|\tilde{y} - y^*\|^2 + \lambda_i \|\tilde{i} - i_{\text{hist}}^{\text{last}}\|_\Delta^2 + \lambda_e \|\tilde{e}\|^2.$$

Terms. The four terms penalize CPI-target miss, GDP-target miss, non-smooth or abrupt FR007 moves, and exchange-rate instability; the weights $\lambda_\pi, \lambda_y, \lambda_i, \lambda_e$ define the targeting rule.

Solution. $\|\cdot\|_\Delta^2$ denotes squared month-to-month FR007 changes with a boundary anchor tying the first counterfactual rate to the last historical FR007. Because paths are linear in $\tilde{\nu}$ and the loss is quadratic, this is a stacked OLS problem solved by one linear-system call in Julia (`A \b`):

Headline Results



- *[to fill in]* The policy-gap magnitude under each targeting rule, including cumulative narrative and HFI shocks required.
- *[to fill in]* Whether counterfactual CPI diverges meaningfully from baseline or remains constrained by weak transmission.
- *[to fill in]* Where the 2019–2020 and 2025 episodes imply similar policy gaps and

Appendix — Asymmetric GDP Gap Specification

Construction. Define the threshold indicator $\mathbb{1}_t^- = \mathbb{1}\{\text{gdp_gap}_t < 1\}$, then split:

$$\text{gap_pos}_t = \text{gdp_gap}_t \cdot (1 - \mathbb{1}_t^-), \qquad \text{gap_neg}_t = \text{gdp_gap}_t \cdot \mathbb{1}_t^- .$$

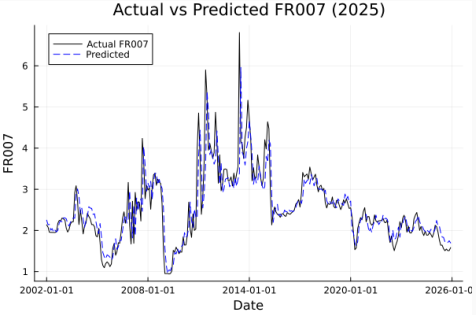
Motivation. Following Chen, Ren & Zha (2018), the PBoC's response to growth is hypothesised to be *asymmetric*: monetary policy reacts more aggressively when realized GDP growth falls meaningfully below the official Government Work Report target than when it merely fluctuates around it.

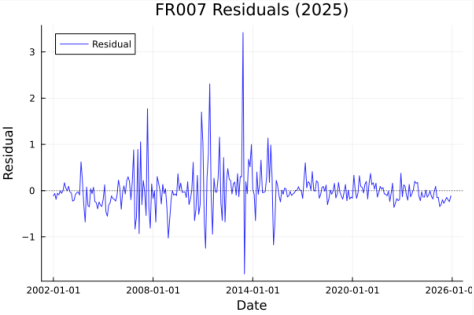
Why a 1pp threshold rather than zero?

- Chinese GWR growth targets are widely understood as floors rather than point estimates — modest overshoots are routine and politically uncontroversial.
- A small undershoot ($0 \leq \text{gap} < 1$) is treated as a near-miss and rarely triggers a policy response.
- Only when the shortfall exceeds roughly one percentage point does the PBoC enter a clearly accommodative regime.

Appendix — Policy Rule Fit and Residuals

Left Actual FR007 vs. predicted FR007 from the rule regression. **Right.** Estimated residual series $\widehat{\varepsilon}_t^{\mathrm{MP}}$, used as the narrative monetary policy shock.





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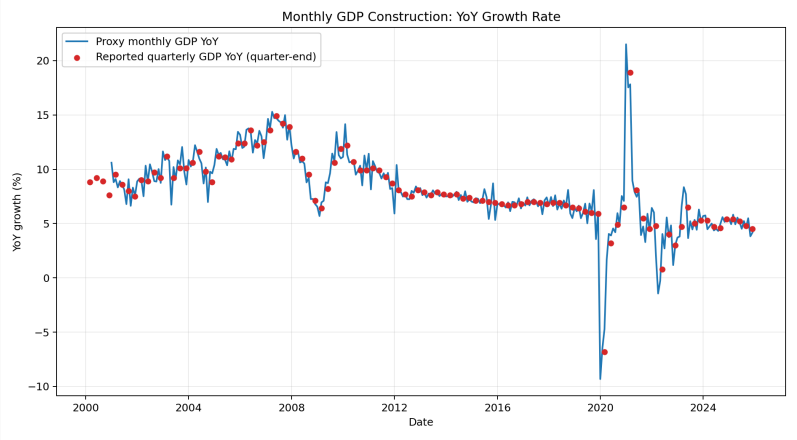
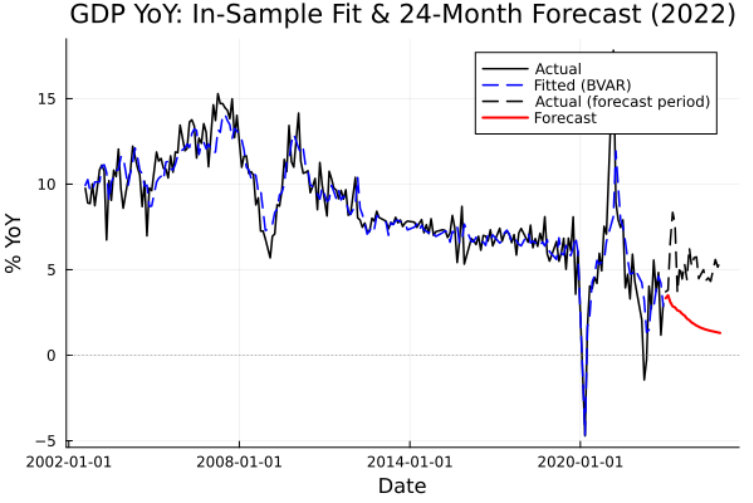
Appendix — Monthly GDP Construction

[To be filled in: Stock & Watson (2010) distribution method, adapted to NBS quarterly GDP (当季值) using the sum identity $Q_T = q_{3T-2} + q_{3T-1} + q_{3T}$, with monthly indicators consumption / FAI / government spending / trade balance.]



Appendix — Forecast vs. Actual (Proxy) Monthly GDP



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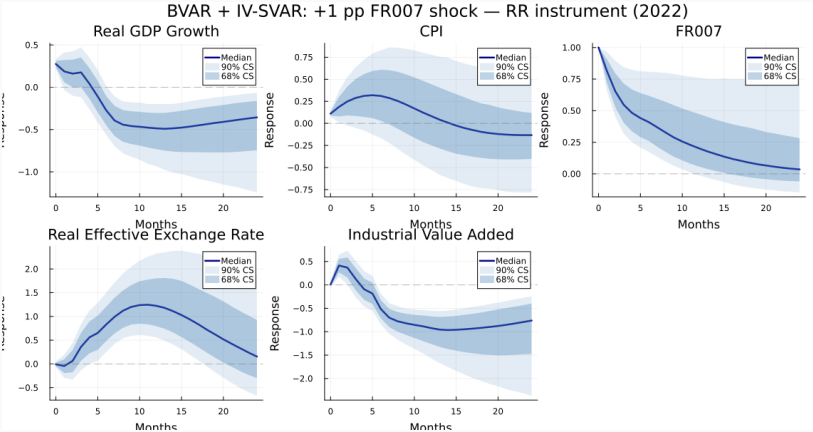
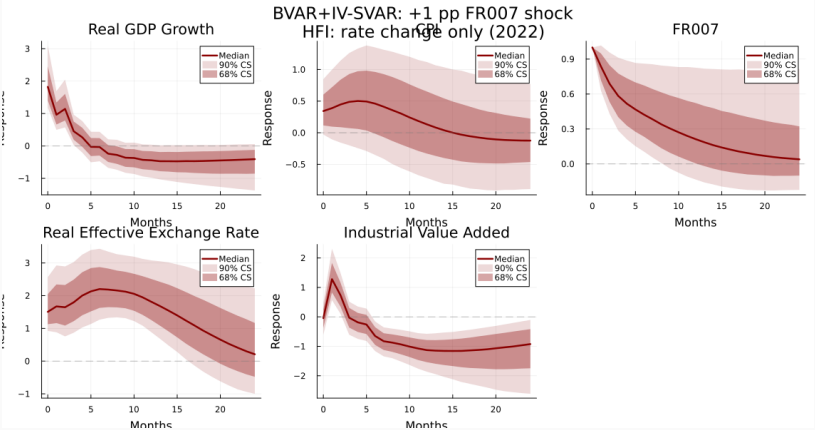
Appendix — Narrative Shock IRFs

Narrative identification. Impulse responses to the estimated monthly policy shock from the rule regression.



Appendix — HFI Shock IRFs

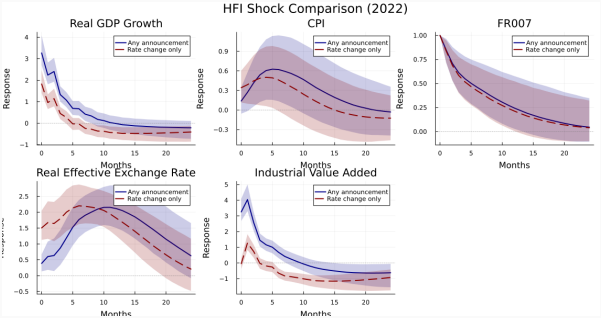
Identification. Impulse responses to the rate-change-only HFI shock series.



The figure reports the median response and uncertainty bands for the main HFI identification scheme.

Appendix — HFI Shocks: IRF Comparison

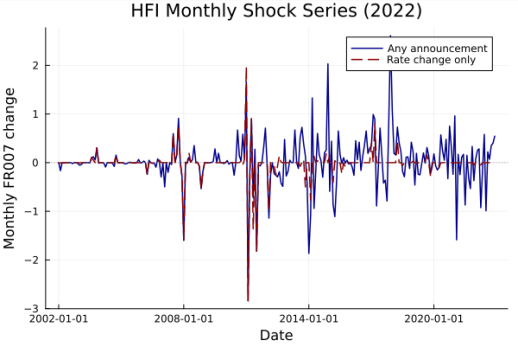
Left. IRF of Chinese economy to monetary policy shocks identified from FR007 surprises on *any* PBoC policy announcement day. **Right.** IRF using *rate-change-only* identification — cleaner shock, smaller sample size.



The rate-change-only series serves as the primary instrument in the counterfactual analysis, with the any-announcement series used for robustness checks.

Appendix — HFI Monthly Shock Series

Monthly aggregation. Daily FR007 surprises are summed within each calendar month, yielding the two monthly HFI shock series used in the counterfactual analysis.



The solid line corresponds to shocks on any policy announcement day; the dashed line uses rate-change-only identification.

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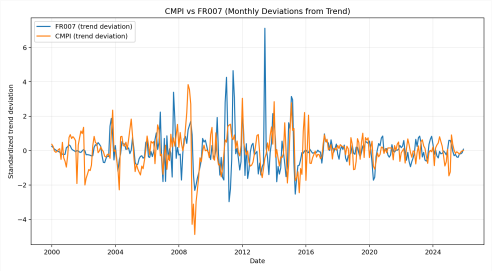
Appendix — CMPI vs. FR007

Specification choice. Comparison of policy rule performance using FR007 (Composite Monetary Policy Indicator our hybrid approach) versus CMPI (Miranda-Agrippino, Nenova & Rey 2026 baseline).



The choice to retain FR007 as the dependent variable while adopting MANR2026's FX specification ensures consistency with the high-frequency identification strategy, which relies on FR007 surprises on policy event dates.

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